

PRISM: Primary-Centric Multimodal Fusion with Information Bottlenecks and Confidence-Aware Gating

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Abstract

Multimodal sentiment analysis predicts continuous or ordinal affect from language, acoustic, and visual cues, but cross-modal alignment and per-stream reliability vary by sample. Previous fusion often collapses modalities into a symmetric or text-fixed representation, which may amplify inherent noise and ignore important information. Therefore, we propose PRISM, a primary-centric framework: (i) each modality is first encoded by a *Variational Token Bottleneck* (VTB), yielding compressed token-level bottleneck representations and confidence estimates; (ii) a *Differentiable Primary Router* (DPR) then selects a sample-dependent primary pathway from these bottleneck representations rather than assuming a fixed dominant modality; and (iii) stacked InfoGate layers refine the selected primary stream with confidence-aware attention and adaptive auxiliary gating. We evaluate the model on both regression and classification datasets. While showing strong performance on regression tasks, the architecture also generalizes to binary humor and sarcasm classification.

1 Introduction

Human affect in real-world communication is expressed through multiple coordinated channels, including lexical content, vocal tone, and visual behavior (Baltrušaitis et al., 2018; Poria et al., 2017). Multimodal sentiment analysis (MSA) therefore aims to infer continuous or discrete sentiment from these heterogeneous signals, and has become a central task in affective computing (Zadeh et al., 2016, 2018; Mai et al., 2022b). Compared with unimodal analysis, MSA exploits complementary cross-modal evidence to achieve more comprehensive emotion understanding (Zadeh et al., 2017; Tsai et al., 2019).

However, multimodal streams are inherently asynchronous, unequally informative, and unequally reliable across samples. Language often

carries explicit sentiment cues, but in some circumstances such as sarcasm, humor, or emotionally charged vocal tones can shift the primary evidence source to other modalities (Castro et al., 2019; Hasan et al., 2019). Most existing fusion pipelines either treat all modalities as symmetric contributors (Zadeh et al., 2017; Liu et al., 2018) or fix language as the dominant rank (Wu et al., 2021; Lei et al., 2023; Zhang et al., 2023a; Xiao et al., 2024), both of which are not able to reach a balance in different modalities.

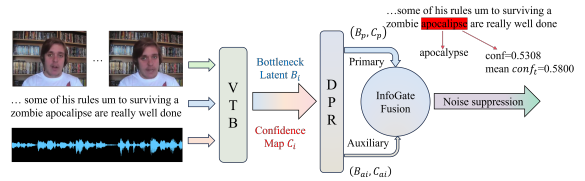


Figure 1: Noise reduction process of PRISM. In this sample, the primary modality is text.

To solve this problem, dynamic primary modality approaches have emerged. HCT-DMG (Wang et al., 2023b) first proposed dynamic primary modality selection. However, HCT-DMG doesn't compress the acoustic and visual modalities which contains more repetitive and irrelevant features, possibly introducing noise interference when treated as primary modality. And HCT-DMG selects primary modality in batch-level instead of sample-level. MODS (Yang et al., 2025) compress acoustic and visual modality with Graph-based Dynamic Compressor (GDC) module, and devises MSelector to select primary modality in sample-level. However, there still exists problems. Firstly, both of them don't filter information in modalities fusion after the selection of primary modality, which might cause information conflict or redundancy. Secondly, the primary modality assignment is a hard discrete decision and is not directly differentiable. The model can optimize the scoring network only indirectly through downstream pre-

diction loss, but it lacks an explicit differentiable training signal for the primary-routing decision itself.

To solve the problem of undirectly back-propagating and lacking explicit noise filtering, we propose PRISM, a primary-centric multimodal fusion framework. First, each modality is independently encoded by a *Variational Token Bottleneck* (VTB) following deep variational IB objectives (Kingma and Welling, 2013; Alemi et al., 2017), producing both a compact latent representation and token-level confidence which will help in later InfoGate module. Second, a *Differentiable Primary Router* (DPR) chooses a sample-dependent primary modality via Gumbel-Softmax (Jang et al., 2017), and the prediction head is routed exclusively through the enhanced primary stream rather than a pooled multimodal vector. Third, stacked InfoGate layers refine the selected primary by attending to auxiliary modalities with confidence-biased cross-modal attention and adaptive gating, whose confidence was generated by former VTB, reducing the risk that unreliable auxiliary tokens distort the final prediction. The noise reduction process is shown in Figure 1.

We first demonstrate the effectiveness of PRISM in the context of MSA. Afterwards, we extend PRISM to classification tasks (including MHD and MSD) (Hasan et al., 2019; Castro et al., 2019) to verify the generalizability of PRISM.

Our main contributions are as follows:

- We devise a module VTB inspired by IB. VTB generate bottlenecks and confidence, which can both compress acoustic and visual information and provide the reliance of tokens.
- DPR adopts a Gumbel-Softmax relaxation to approximate differentiable primary-modality selection. This design allows gradients from the task objective to flow directly into the routing module, improving its trainability.
- Compared to other dynamic modality selection approach, we calculate each token’s reliance with confidence in fusion stage, to explicitly extract useful information.

2 Related Work

Fixed Modality. Early MSA research treated all modalities as symmetric contributors. MulT (Tsai et al., 2019) employs pairwise cross-modal attention to align asynchronous multimodal sequences;

MISA (Hazarika et al., 2020) projects each modality into invariant and specific subspaces; and Hy-Con (Mai et al., 2022b) uses hybrid contrastive learning for tri-modal representations. Later work shifted toward language-centric static primary: TCSP (Wu et al., 2021) distinguishes shared and private semantics anchored on text, and several Transformer-based methods (Lei et al., 2023; Zhang et al., 2023a; Wang et al., 2023a, 2025) inject non-verbal cues into language as the dominant stream.

But these symmetric or fixed-primary designs have inherent limitations. Symmetric fusion dilutes the most informative modality with noisy auxiliary streams, while language-centric methods rigidly anchor on text even when acoustic or visual cues carry the decisive evidence—in sarcasm, humor, or emotionally charged speech, the primary modality varies by sample. Moreover, none of these approaches estimate per-token reliability before fusion.

Dynamic Modality Selecting. To overcome the rigidity of fixed-primary pipelines, dynamic primary selection has emerged. HCT-DMG (Wang et al., 2023b) first proposed dynamic modality gating, selecting a primary modality per batch without compressing acoustic or visual features. MODS (Yang et al., 2025) advances to sample-level selection via MSelector and employs a Graph-based Dynamic Compressor (GDC) to condense acoustic and visual sequences. EMOE (Fang et al., 2025) explores soft expert mixing with entropy balancing.

However, these methods operate on pooled modality representations after global aggregation: they do not estimate which individual *tokens* within a modality carry reliable information, nor do they modulate fusion weights at the token level based on per-element confidence. Without token-level filtering, noisy segments within an otherwise informative modality can still corrupt the fused representation.

Information Bottleneck. The Information Bottleneck (IB) principle (Tishby et al., 2000) provides a framework for compressing representations while preserving task-relevant information. Deep variational IB (Alemi et al., 2017) enables stochastic neural encoders. In multimodal learning, MultiModal-IB (Mai et al., 2022a) jointly compresses unimodal and multimodal representations; ITHP (Xiao et al., 2024) chains multiple hierarchical bottlenecks with information-theoretic objec-

tives; CyIN (Lin et al., 2025) constructs an informative latent space via token- and label-level IB with cyclic cross-modal interaction.

However, existing IB methods either treat compression as a standalone regularizer decoupled from downstream fusion decisions, or chain bottlenecks over a fixed primary modality without per-sample adaptation. None of them export a token-level confidence signal that can directly control how auxiliary information is gated into the primary stream. In contrast, PRISM’s VTB instantiates per-modality variational bottlenecks in parallel at token resolution, producing per-token confidence that drives both primary routing and InfoGate attention.

3 Methodology

3.1 Problem Formulation and Overview

Given an utterance-level video segment, our goal is to predict its sentiment intensity by leveraging multimodal sequential data. The input consists of language, acoustic, and visual modalities, denoted as $\mathcal{M} = \{t, a, v\}$. For each modality $m \in \mathcal{M}_{\text{reg}}$, we denote its raw feature sequence as $\mathbf{X}_m \in \mathbb{R}^{T_m \times d_m}$, where T_m is the sequence length and d_m is the original feature dimension. We denote the prediction head output as $\hat{y}$, which is supervised by the ground-truth label y .

Unlike fusion paradigms that treat all modalities equally or default to language as the primary trunk, PRISM dynamically routes the most reliable modality for each sample as the primary stream, leveraging the remaining modalities as auxiliary evidence to prevent noise from propagating into the final prediction. This primary-centric design contrasts with two common alternatives: symmetric fusion (e.g., MulT (Tsai et al., 2019), MISA (Hazari et al., 2020)) where all modalities attend to one another and noise can cross-contaminate, and fixed-text pipelines (e.g., MAG-BERT (Rahman et al., 2020)) that fail when language is not the dominant evidence carrier. By designating a single prediction pathway and treating all other modalities as optional supplements, PRISM protects the final representation from unreliable streams.

As illustrated in Figure 2 and Algorithm 1, PRISM operates in four sequential stages: (i) encoding each modality with a *Variational Token Bottleneck* (VTB) to extract latents and token-level confidence estimates; (ii) employing DPR for sample-wise primary modality routing; (iii) refin-

Algorithm 1 PRISM forward pass on a batch.

- 1: Encode modality-specific features $\mathbf{F}_m \rightarrow (\mathbf{B}_m, \mathbf{C}_m)$ for $m \in \mathcal{M}$.
 - 2: Compute token-level and label-level IB objectives: $\mathcal{L}_{\text{tib}}$ and $\mathcal{L}_{\text{lib}}$.
 - 3: Obtain routing weights and hard selection $(w, r) \leftarrow \text{DPR}(\{\mathbf{B}_m\}_{m \in \mathcal{M}})$.
 - 4: Route primary and auxiliary bottlenecks $(\mathbf{B}_p, \mathbf{C}_p, \mathcal{A}) \leftarrow \text{Route}(\mathbf{B}_m, \mathbf{C}_m, w, r)$.
 - 5: Refine primary representation via InfoGate $\mathbf{B}_p^{\text{out}} \leftarrow \text{InfoGateStack}(\mathbf{B}_p, \mathbf{C}_p, \mathcal{A})$.
 - 6: Predict output $\hat{y} \leftarrow \text{Head}(\text{Pool}(\mathbf{B}_p^{\text{out}}))$.
 - 7: $\mathcal{L}_{\text{rib}} \leftarrow 0$.
 - 8: **if** $s = 2$ **then**
 - 9: Update routing supervision $\mathcal{L}_{\text{rib}}$.
 - 10: **end if**
 - 11: **return** $\hat{y}, \alpha_{\text{IB}}(\mathcal{L}_{\text{tib}} + \mathcal{L}_{\text{lib}}) + \lambda_{\text{rib}}\mathcal{L}_{\text{rib}}$.
-

ing the primary representation via *InfoGate* layers, which use confidence-aware attention and gating to asymmetrically fuse auxiliary signals; and (iv) predicting the target exclusively from the enhanced primary representation.

3.2 VTB Representation Learning

Let $\mathbf{F}_m \in \mathbb{R}^{N \times T \times d}$ denote the projected features for modality $m \in \mathcal{M}_{\text{reg}} = \{t, a, v\}$. d_b denote the bottleneck width. To reduce modality-specific redundancy and task-irrelevant variation before fusion, each stream is encoded by a *Variational Token Bottleneck* (VTB) inspired by *Information Bottleneck* (IB) (Tishby et al., 2000; Alemi et al., 2017), predicting a Gaussian mean and log-variance:

$$[\boldsymbol{\mu}_m; \log \boldsymbol{\sigma}_m^2] = \text{VTB}_m(\mathbf{F}_m), \quad (1)$$

yielding $\boldsymbol{\mu}_m, \log \boldsymbol{\sigma}_m^2 \in \mathbb{R}^{N \times T \times d_b}$. During training, representations are sampled using the reparameterization trick (Kingma and Welling, 2013):

$$\mathbf{B}_m = \boldsymbol{\mu}_m + \boldsymbol{\sigma}_m \odot \boldsymbol{\epsilon}, \quad \mathbf{B}_m \in \mathbb{R}^{N \times T \times d_b}, \quad (2)$$

where $\boldsymbol{\sigma}_m = \exp(\frac{1}{2} \log \boldsymbol{\sigma}_m^2)$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ has the same shape as $\mathbf{B}_m$, and $\odot$ denotes element-wise multiplication. Crucially, the estimated log-variance reflects encoding uncertainty; we convert it into a *confidence map* aligned with the bottleneck tokens:

$$\mathbf{C}_m = \sigma(-\log \boldsymbol{\sigma}_m^2), \quad \mathbf{C}_m \in [0, 1]^{N \times T \times d_b}, \quad (3)$$

with $\sigma(\cdot)$ applied element-wise. Equivalently, $\mathbf{C}_m$ has the same shape as $\mathbf{B}_m$ and assigns higher values to lower-variance bottleneck dimensions. This

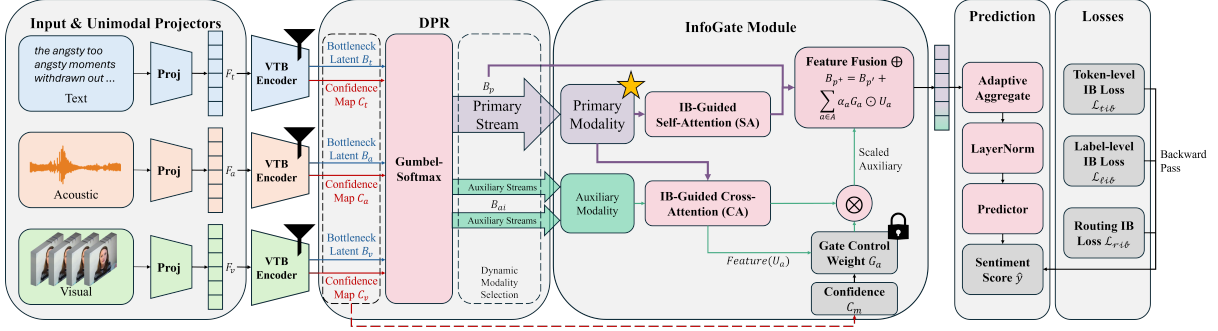


Figure 2: Architecture of PRISM under the MSA regression setting. Each modality is first compressed through a Variational Token Bottleneck (VTB) that yields a bottleneck representation with a token-level confidence map; the primary modality is then selected by DPR via Gumbel–Softmax routing; stacked InfoGate layers asymmetrically fuse auxiliary modalities through confidence-modulated cross-attention and adaptive gating. The final prediction is produced from the enhanced primary representation.

confidence map is later used by InfoGate to bias cross-modal attention scores and gate auxiliary values (Sec. 3.4). The variational formulation simultaneously yields a compact latent and an uncertainty estimate—the variance directly reflects how reliably each token is encoded. While recent IB-based methods such as Correlation IB (Jiang et al., 2024) applies bottlenecks over fused or distilled representations, PRISM’s parallel per-modality VTB preserves modality-specific variance structures and exports per-token confidence, enabling the downstream routing and gating that purely representation-level IB methods cannot support. We encode modalities in parallel rather than chaining bottlenecks over a fixed primary (as in ITHP (Xiao et al., 2024)), therefore, preserving per-modality characteristics and preventing error cascades across streams.

Token-level IB objective. To regularize the bottleneck space, we define a token-level objective balancing compression with reconstruction:

$$\mathcal{L}_{\text{tib}} = \sum_{m \in \mathcal{M}} \left[\text{KL}(q(\mathbf{B}_m | \mathbf{F}_m) \| \mathcal{N}(\mathbf{0}, \mathbf{I})) + \beta_{\text{IB}} \mathbb{E} \|g_m(\mathbf{B}_m) - \mathbf{F}_m\|_2^2 \right], \quad (4)$$

where g_m is a modality-specific decoder. The second term is a token-level *self-reconstruction* loss: it encourages the sampled bottleneck $\mathbf{B}_m$ to retain enough information to recover the projected features $\mathbf{F}_m$, while the KL term penalizes excessive latent capacity. Together they instantiate the standard IB trade-off between compression and fidelity (Aleml et al., 2017).

Label-level IB objective. To retain task-relevant semantics, a label-level objective applies a probe

ψ_m on the mean-pooled bottleneck $\bar{\mathbf{B}}_m$:

$$\mathcal{L}_{\text{lib}} = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \left[\text{KL}(q(\bar{\mathbf{B}}_m) \| \mathcal{N}(\mathbf{0}, \mathbf{I})) + \beta_{\text{IB}} e_m \right], \quad (5)$$

where $e_m = |\psi_m(\bar{\mathbf{B}}_m) - y|$ evaluates the modality-specific prediction error.

3.3 Differentiable Primary Router (DPR)

To adapt to varying sample-level informativeness (Yang et al., 2025), DPR designates a single primary stream per input. We first compute an attention-based summary for each modality:

$$\mathbf{h}_m = \text{softmax} \left(\frac{\mathbf{B}_m \mathbf{w}_m}{\sqrt{d_b}} \right)^\top \mathbf{B}_m, \quad (6)$$

where $\mathbf{w}_m \in \mathbb{R}^{d_b}$ is a modality-specific learnable query that scores each bottleneck token before pooling. Routing operates on VTB-compressed sequences rather than raw features, so the router bases its decision on task-filtered information. The routing logits are obtained by

$$\mathbf{z} = f_{\text{route}}(\text{concat}_{m \in \mathcal{M}}(\mathbf{h}_m)), \quad (7)$$

where $f_{\text{route}}(\cdot)$ is a two-layer MLP over the modality summaries $\{\mathbf{h}_m\}_{m \in \mathcal{M}}$ (each $\mathbf{h}_m \in \mathbb{R}^{d_b}$). The summaries are used only for routing and are not fed to InfoGate. Then we use Gumbel-Softmax (Jang et al., 2017) to produce a hard primary selection. It is used to obtain a differentiable hard primary indicator r and soft routing weights w :

$$r = \text{GumbelSoftmax}(\mathbf{z}, \tau_{\text{gumbel}}), \quad (8)$$

$$w = \text{softmax}(\mathbf{z} / \tau_{\text{gumbel}}). \quad (9)$$

Here, $\mathbf{r} = (r_1, \dots, r_M)^\top$ is a one-hot routing vector over the M candidate modalities, where $r_m \in \{0, 1\}$ and $\sum_{m \in \mathcal{M}} r_m = 1$. Thus, $r_m = 1$ indicates that modality m is selected as the primary modality for the current sample, while $r_m = 0$ means that modality is treated as auxiliary. In contrast, $\mathbf{w} = (w_1, \dots, w_M)^\top$ is the relaxed soft routing distribution with $w_m \in (0, 1)$ and $\sum_m w_m = 1$, which is used for differentiable training and routing supervision. From the hard indicator r , we assemble the primary and auxiliary *token sequences* as

$$\mathbf{B}_p = \sum_m r_m \mathbf{B}_m, \quad (10)$$

along with its confidence $\mathbf{C}_p$. Modalities where $r_m = 0$ are routed as the auxiliary set $\mathcal{A}$. Unlike MODS’s MSelector (Yang et al., 2025), which determines the primary modality by an $\arg \max$ over softmax scores and therefore cannot back-propagate through the discrete primary-selection operation, DPR uses a straight-through Gumbel–Softmax router on VTB-filtered bottlenecks. The forward pass still commits to a single sample-wise primary stream, but the backward pass propagates gradients through the relaxed routing weights, making the primary designation itself trainable. The selected primary then remains the prediction pathway throughout subsequent InfoGate layers, preserving a clear primary–auxiliary asymmetry.

3.4 Confidence-Aware InfoGate Fusion

After DPR selects the primary stream $\mathbf{B}_p$, the remaining modalities serve as auxiliary evidence. However, even an informative auxiliary modality may contain low-confidence or task-irrelevant token positions. InfoGate therefore performs *asymmetric* auxiliary-to-primary fusion: auxiliary streams refine $\mathbf{B}_p$ through confidence-modulated cross-attention and adaptive gating, while the prediction head reads only the enhanced primary representation. This differs from symmetric cross-modal attention in multimodal Transformers (Tsai et al., 2019; Lv et al., 2021; Wang et al., 2023b), where all modalities exchange information uniformly and unreliable auxiliary evidence can propagate across streams.

For each auxiliary stream $a \in \mathcal{A}$, InfoGate applies an auxiliary-to-primary cross-attention block CA_a . Let $\mathbf{Q} = \mathbf{W}_q \mathbf{B}_p$, $\mathbf{K} = \mathbf{W}_k \mathbf{B}_a$, and $\mathbf{V} = \mathbf{W}_v \mathbf{B}_a$ denote the query, key, and value projections, respectively. Here $\mathbf{Q}$ comes from the primary stream, while $\mathbf{K}$ and $\mathbf{V}$ come from auxiliary

stream a . We denote the pre-softmax attention logit between primary query position t and auxiliary key position j as

$$A_{t,j} = \frac{\mathbf{Q}_t^\top \mathbf{K}_j}{\sqrt{d_h}}. \quad (11)$$

Using the VTB confidence map $\mathbf{C}_a$ of auxiliary stream a , we compute a scalar confidence for auxiliary position j :

$$\bar{c}_{a,j} = \frac{1}{d_b} \sum_{k=1}^{d_b} [\mathbf{C}_a]_{j,k}. \quad (12)$$

This confidence modulates both the pre-softmax logits and the value vectors:

$$\tilde{A}_{t,j} = A_{t,j} + \gamma \log(\bar{c}_{a,j} + \epsilon), \quad \tilde{\mathbf{V}}_j = \bar{c}_{a,j} \mathbf{V}_j, \quad (13)$$

where γ is a learnable scaling factor and $\epsilon > 0$ avoids $\log 0$. The log-confidence term lowers the attention probability assigned to uncertain auxiliary keys, while value scaling reduces how much low-confidence auxiliary content contributes to the attended output. The resulting auxiliary context is

$$[\mathbf{U}_a]_t = \sum_j \text{softmax}_j(\tilde{A}_{t,\cdot}) \tilde{\mathbf{V}}_j, \quad (14)$$

which corresponds to

$$\mathbf{U}_a = \text{CA}_a(\mathbf{B}_p, \mathbf{B}_a, \mathbf{C}_a), \quad a \in \mathcal{A}. \quad (15)$$

Here $\mathbf{U}_a$ denotes the auxiliary-to-primary context that transfers complementary evidence from stream a into the primary pathway at each time step.

While confidence-modulated attention controls *which* auxiliary tokens contribute to $\mathbf{U}_a$, an adaptive gate further regulates *how much* of this auxiliary context is admitted into the primary stream (Xie et al., 2024):

$$\mathbf{G}_a = \sigma(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 [\mathbf{B}'_p; \mathbf{U}_a])), \quad (16)$$

with $\mathbf{B}'_p = \mathbf{B}_p + \text{SA}(\mathbf{B}_p)$ representing the self-attended primary sequence. The primary representation updates via:

$$\mathbf{B}_p^+ = \mathbf{B}'_p + \sum_{a \in \mathcal{A}} \alpha_a \mathbf{G}_a \odot \mathbf{U}_a, \quad (17)$$

$$\alpha_a = \max\left(\eta_{\min}, \frac{1 + \cos(\bar{\mathbf{B}}_p, \bar{\mathbf{B}}_a)}{2}\right). \quad (18)$$

The coefficient α_a provides a residual safeguard: $\eta_{\min}$ sets a minimum bound on auxiliary information flow even during temporary representational divergence, and is empirically chosen by performing

multiple experiments, usually set as 0.3. This dual control—logit/value confidence modulation plus adaptive gating—differs from confidence-aware methods that apply a single scalar per modality (Li et al., 2024); InfoGate applies token-level modulation to auxiliary-to-primary fusion.

3.5 Training Objective

PRISM is trained with task prediction, IB regularization, and routing supervision. During warm-up, routing supervision is disabled by setting $\lambda_{\text{rib}} = 0$.

Routing supervision. Soft routing targets are derived from the label-level probe errors:

$$t_m = \frac{\exp(-e_m/\tau_{\text{sel}})}{\sum_{m' \in \mathcal{M}} \exp(-e_{m'}/\tau_{\text{sel}})}. \quad (19)$$

For samples $\mathcal{S}$ whose modality-error gap exceeds a margin δ , the routing loss is

$$\mathcal{L}_{\text{rib}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \text{KL}(w^{(i)} \parallel t^{(i)}). \quad (20)$$

Task loss. Let N be the mini-batch size. For MSA regression, we devise an loss that combines L1 and MSE:

$$\mathcal{L}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^N (|\hat{y}_i - y_i| + \rho_{\text{mse}}(\hat{y}_i - y_i)^2), \quad (21)$$

where the L1 term matches MAE-style sentiment evaluation and the MSE term penalizes large errors. Here ρ_{mse} weights the MSE term

For binary humor and sarcasm detection, the task loss is binary cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]. \quad (22)$$

Thus the task term is selected according to the task:

$$\mathcal{L}_{\text{task}} = \begin{cases} \mathcal{L}_{\text{reg}}, & \text{regression,} \\ \mathcal{L}_{\text{cls}}, & \text{classification.} \end{cases} \quad (23)$$

Overall objective. The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha_{\text{IB}}(\mathcal{L}_{\text{tib}} + \mathcal{L}_{\text{lib}}) + \lambda_{\text{rib}}\mathcal{L}_{\text{rib}}, \quad (24)$$

where α_{IB} weights the IB regularizers, and λ_{rib} weights routing supervision.

4 Experiments

Table 1 reports CMU-MOSI and CMU-MOSEI (Zadeh et al., 2016, 2018). Table 2 reports CH-SIMS v2 (Liu et al., 2022), and Figure 3 reports UR-FUNNY v2 (Hasan et al., 2019) and MUSTARD (Castro et al., 2019). PRISM entries use the official train/dev/test splits; test numbers are reported at the checkpoint chosen on the development set (Section 4.2), following standard benchmarks.

4.1 Datasets and Evaluation Metrics

4.1.1 MSA

We evaluate PRISM on CMU-MOSI, CMU-MOSEI, and CH-SIMS v2. We report Acc-7, Acc-2, F1-score, Mean Absolute Error (MAE), and Pearson Correlation (Corr) on CMU-MOSI and CMU-MOSEI, and replace Acc7 with Acc-5 and Acc-3 on CH-SIMS v2.

4.1.2 MSD and MHD

We evaluate PRISM on UR-FUNNY v2 and MUSTARD. We report binary classification accuracy under the HKT-aligned setting with HCF as an additional modality (Hasan et al., 2021).

4.2 Implementation Details

4.2.1 Backbones

For MOSI and MOSEI we use a DeBERTa-v3-base (He et al., 2021) text encoder while for CH-SIMS v2 (Liu et al., 2022) we use BERT-base-chinese (Devlin et al., 2019). We represent acoustic and visual modality with lightweight audio and visual projectors, and for classification we follow the HKT-aligned convention with an ALBERT-base-v2 (Lan et al., 2020) text encoder and the additional HCF modality processed through its own projector and VTB. All modalities are mapped to a shared hidden size and passed through modality-specific VTB encoders that expose a bottleneck sequence $\mathbf{B}_m$ and a confidence map $\mathbf{C}_m$ (Section 3.2).

4.2.2 Optimization

We optimize with AdamW using the two-stage objective described in Section 3.5. Checkpoints for regression are chosen by development-set MAE; for classification, by development-set accuracy. Hyperparameters are determined based on the development set. Dataset-specific numerical settings are summarized in Appendix A.

Table 1: Comparison with baselines on CMU-MOSI and CMU-MOSEI. Acc-2 and F1 are reported under the non-zero protocol, where neutral samples with zero sentiment scores are excluded. Results marked with $\dagger$ are from the original papers, and the remaining results are reproduced by us. Best and second-best results are shown in bold and underlined, respectively (Zadeh et al., 2016, 2018).

Baseline Models	CMU-MOSI					CMU-MOSEI				
	Acc7 $\uparrow$	Acc2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$	Acc7 $\uparrow$	Acc2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$
DLF $\dagger$ (Wang et al., 2025)	47.1	85.1	85.0	0.731	0.781	53.9	85.4	85.3	0.536	0.764
KuDA $\dagger$ (Feng et al., 2024)	47.1	86.4	86.5	0.705	0.795	52.9	86.5	86.6	0.529	0.776
DEVA $\dagger$ (Wu et al., 2025)	46.3	86.3	86.3	0.730	0.787	52.3	86.1	86.2	0.541	0.769
C-MIB (Mai et al., 2022a)	47.7	87.8	87.8	0.662	0.835	52.7	86.9	86.8	0.542	0.784
ALMT $\dagger$ (Zhang et al., 2023a)	49.4	86.4	86.4	0.683	0.805	54.2	86.8	86.9	0.526	0.779
ITHP (Xiao et al., 2024)	47.7	<u>88.5</u>	<u>88.5</u>	0.663	0.856	52.2	<u>87.1</u>	<u>87.1</u>	0.550	<u>0.792</u>
Multimodal Boosting (Mai et al., 2023a)	49.1	<u>88.5</u>	<u>88.4</u>	<u>0.634</u>	<u>0.855</u>	54.0	86.5	86.5	<u>0.523</u>	0.779
DTN (Zeng et al., 2024)	48.1	86.2	86.2	0.714	0.807	52.5	86.3	86.3	0.579	<u>0.792</u>
DMD (Li et al., 2023)	44.9	84.3	84.3	0.726	0.788	52.8	84.6	84.6	0.538	0.768
DHMD $\dagger$ (Li et al., 2026)	<u>49.9</u>	87.8	87.7	0.683	0.812	<u>55.3</u>	<u>87.1</u>	87.0	0.527	0.786
EMOE (Fang et al., 2025)	45.2	84.8	84.8	0.723	0.790	52.5	85.0	85.0	0.542	0.760
MODS $\dagger$ (Yang et al., 2025)	49.3	85.8	86.0	0.688	0.798	54.3	85.9	86.1	0.527	0.772
PRISM	50.8	89.0	89.0	0.606	0.856	56.0	87.4	87.3	0.499	0.800

Table 2: Comparison with baselines on CH-SIMS v2 dataset (Liu et al., 2022).

Baseline Models	CH-SIMS v2					
	Acc5 $\uparrow$	Acc3 $\uparrow$	Acc2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$
LF-DNN (Williams et al., 2018)	51.8	71.2	77.8	77.9	0.322	0.668
TFN (Zadeh et al., 2017)	53.3	70.9	78.1	78.1	0.322	0.662
LMF (Liu et al., 2018)	51.6	70.0	77.8	77.8	0.327	0.651
MISA (Hazarika et al., 2020)	47.5	68.9	78.2	78.3	0.342	0.671
MAG-BERT (Rahman et al., 2020)	49.2	70.6	77.1	77.1	0.346	0.641
Self-MM (Yu et al., 2021)	53.5	72.7	78.7	78.6	0.315	0.691
MMIM (Han et al., 2021)	50.5	70.4	77.8	77.8	0.339	0.641
DLF (Wang et al., 2025)	47.5	70.0	78.1	77.9	0.346	0.683
TMDC (Zhuang et al., 2026)	53.8	71.9	<u>79.6</u>	<u>79.6</u>	0.356	0.703
Diffusion Bridge (Lee et al., 2025)	52.5	70.7	78.6	79.0	0.323	0.677
MODS (Yang et al., 2025)	<u>55.5</u>	73.7	<u>79.6</u>	79.5	<u>0.297</u>	0.712
PRISM	55.6	<u>73.4</u>	80.1	80.1	0.291	<u>0.705</u>

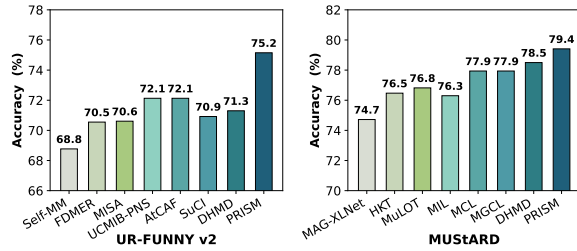


Figure 3: Comparison with classification baselines on UR-FUNNY v2 and MUSTARD (accuracy, %) (Hasan et al., 2019; Castro et al., 2019).

4.3 Comparison with State-of-the-art Methods

Tables 1 and 2, together with Figure 3, report the comparison results on MSA regression and affective classification benchmarks. PRISM obtains State-of-the-art performance, showing that dynamically selecting a primary stream and refining it with confidence-aware auxiliary evidence is effective for multimodal sentiment regression.

Compared with symmetric fusion methods, fixed

Table 3: Ablation study results of PRISM components on CMU-MOSEI and CH-SIMS v2 datasets.

Setting	CMU-MOSEI			CH-SIMS v2		
	MAE $\downarrow$	Acc7 $\uparrow$	Acc2 $\uparrow$	MAE $\downarrow$	Acc5 $\uparrow$	Acc2 $\uparrow$
PRISM (full)	0.499	56.0	87.4	0.291	55.6	80.1
w/o VTB	0.501	55.5	86.4	0.302	53.8	80.3
w/o DPR	0.510	55.3	86.9	0.322	53.5	77.2
w/o InfoGate	0.509	54.2	87.2	0.345	51.5	77.1

text-centered models and even some earlier Dynamic modality selection frameworks, PRISM uses DPR to adapt the primary modality at the sample level, while InfoGate filters auxiliary information with IB-derived confidence before injecting it into the primary stream. This design explicitly reduces the influence of unreliable modalities and leads to more robust fused representations. On UR-FUNNY v2 (Hasan et al., 2019), PRISM (75.2%) surpasses earlier fusion baselines including UCMIB-PNS (Chen et al., 2025) and AtCAF (Huang et al., 2025) (72.1%); on MUSTARD (Castro et al., 2019), it reaches 79.4%, competitive with DHMD (78.5%). These results indicate that the same primary-centric framework generalizes to MHD/MSD.

4.4 Ablation Study

To further verify the effectiveness of the proposed components, we conduct ablation experiments on CMU-MOSEI and CH-SIMS v2, as shown in Table 3. All variants use the same hyper-parameters within each dataset.

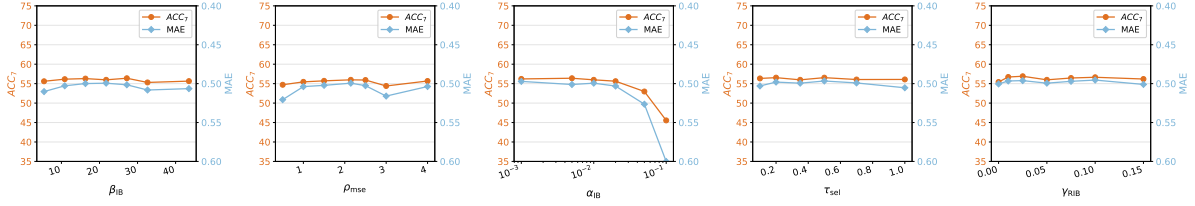


Figure 4: Hyper-parameter sensitivity on CMU-MOSEI. Each panel varies one quantity on the horizontal axis (β_{IB} , α_{IB} , ρ_{mse} , τ_{sel} , or λ_{rib}); all other hyper-parameters match Appendix A. Orange circles: ACC_7 (%); blue diamonds: MAE (axis inverted so lower error reads toward the top).

4.4.1 Effectiveness of VTB

To evaluate VTB, we replace each stochastic bottleneck encoder with a deterministic linear projection, thereby removing the information bottleneck constraint before fusion. The performance degradation indicates that directly passing modality features into the fusion module makes the model more susceptible to modality-specific redundancy and noise. This confirms that VTB provides a useful pre-fusion compression mechanism that preserves task-relevant information while suppressing less reliable details.

4.4.2 Importance of DPR

To validate DPR, we disable dynamic primary routing and use text as the fixed primary modality for all samples. The results show that static primary assignment weakens the model, suggesting that a single fixed modality cannot consistently serve as the most reliable prediction anchor across diverse samples. This demonstrates the importance of sample-level primary selection, especially when nonverbal cues provide complementary or contradictory affective evidence.

4.4.3 Effect of InfoGate

To examine the role of InfoGate, we remove the full InfoGate fusion stack and replace it with a simpler cross-modal fusion path. The resulting degradation shows that merely aggregating multimodal features is insufficient for robust sentiment prediction. InfoGate enables the selected primary representation to be selectively enriched by auxiliary modalities, reducing unreliable cross-modal injection and improving fine-grained affective modeling.

4.4.4 Sensitivity analysis

We assess robustness on CMU-MOSEI by varying *one* scalar hyper-parameter at a time while freezing all remaining settings to the development-selected configuration. We sweep β_{IB} , α_{IB} , ρ_{mse} ,

τ_{sel} , and λ_{rib} , which govern the IB terms, the MSE component of the regression task objective, the selector target temperature, and the routing supervision scale in Section 3.5. Figure 4 reports test ACC_7 and MAE with dual vertical axes (higher ACC_7 and lower MAE are preferable). Metrics stay within a moderate band over the explored grids, indicating that PRISM is not overly sensitive to these knobs around the chosen optimum.

4.5 Qualitative Evaluation

We illustrate PRISM on one CH-SIMS v2 sample which shows cross-modal conflict. In Figure 5, language conveys negativity (nobody comforts me), but audio-visual cues suggest positivity. The character smiles and speaks with a relieved voice. PRISM assigns high weight and high confidence to visual and acoustic input, correctly predicting positive sentiment.

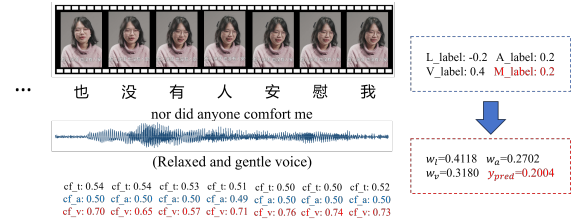


Figure 5: Display of cases and modality weights on CH-SIMS v2. Only part of the sample is shown.

5 Conclusion

We propose PRISM, a primary-centric framework that compresses each modality with a Variational Token Bottleneck, dynamically selects a sample-dependent primary pathway, and enhances it through confidence-aware InfoGate fusion. Experiments on sentiment regression, humor detection, and sarcasm detection benchmarks demonstrate the effectiveness of our work.

Limitations

PRISM’s confidence estimation and routing supervision introduce additional training-time objectives, which may increase optimization cost when scaling to larger multimodal backbones. These auxiliary losses are not computed during inference, so the added cost is mainly associated with training rather than deployment-time loss evaluation. Future work could investigate more lightweight supervision or distillation strategies for scaling PRISM to larger multimodal models.

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A Hyper-parameter Settings	867
We select hyper-parameters on the development set using a lightweight two-stage Optuna search (Akiba et al., 2019): an initial random exploration stage followed by TPE-based refinement over promising regions. The search is restricted to the main training and architectural parameters, including batch	868

size, learning rates, dropout, hidden dimensionality, bottleneck dimensionality, the number of InfoGate layers, attention heads, and the weights of the auxiliary information-bottleneck and routing losses. Other implementation choices, such as the optimizer, warmup schedule, maximum sequence length, and checkpoint-selection rule, are kept fixed across runs. For regression datasets, checkpoints are selected by development-set MAE; for classification datasets, they are selected by development-set accuracy.

Table 4 summarizes the broad ranges used during development. For the sensitivity analysis in Figure 4, we vary one scalar hyper-parameter at a time around the development-selected configuration while keeping all remaining settings fixed. The swept quantities (β_{IB} , α_{IB} , ρ_{mse} , τ_{sel} , and λ_{rib}) are chosen from the same ranges as the development-set search.

Table 4: Compact hyper-parameter search space.

Hyper-parameter	Search range / candidates
Train batch size	{4, 8, 16}
Effective batch size	{16, 32, 64}
Backbone learning rate	$[8 \times 10^{-6}, 4 \times 10^{-5}]$
Fusion learning rate	$[2 \times 10^{-4}, 2 \times 10^{-3}]$
Dropout rate	[0.05, 0.35]
Shared hidden dimension	{128, 256, 384}
Bottleneck dimension	{64, 96, 128, 192}
InfoGate layers	{2, 3, 4, 5}
Attention heads	{4, 8}
α_{IB}	$[10^{-4}, 10^{-2}]$
β_{IB}	[5, 25]
ρ_{mse}	[0.5, 2.5]
τ_{sel}	[0.3, 0.9]
λ_{rib}	[0.04, 0.15]

B Model Complexity Analysis

Table 5 reports the computational footprint of PRISM on CMU-MOSI and CMU-MOSEI. FLOPs are measured for forward passes only. Specifically, “FLOPs / train fwd” denotes the forward cost of a typical training micro-batch. “Train fwd FLOPs / epoch” reports only training-batch forward passes. The table also reports the per-epoch forward costs for development and test evaluation separately. Peak memory and per-epoch time are measured from the full training loop on a single NVIDIA GeForce RTX 4090 D GPU with the same micro-batch and gradient accumulation settings as the paper configurations.

Table 5: Model complexity metrics for MOSI and MOSEI. FLOPs are measured for forward passes only.

	MOSI	MOSEI
Parameters	191.8M	185.2M
FLOPs / train fwd	89.79G	26.69G
Train fwd FLOPs / epoch	7.20T	108.51T
Dev eval FLOPs / epoch	1.20T	9.19T
Test eval FLOPs / epoch	3.58T	22.82T
Peak train memory (GiB)	8.958	5.715
Avg. epoch time (full loop, s)	14.68	517.04
Batches / epoch	81	4,067
Batch $\times$ accum.	16×2	4×8

C Loss Component Learning Curves

Figure 6 shows the learning curves of the main loss components on CMU-MOSI (Zadeh et al., 2016). Each curve is averaged over every ten training iterations to make the optimization trend smoother. Firstly, it can be seen from Fig. 6 (a) that the total loss decreases substantially after the early training phase and continues to drop gradually afterwards. This indicates that the combination of the prediction objective and auxiliary regularizers can be optimized stably. For the task loss in Fig. 6(b), the curve follows a similar downward trend and converges to a small value, suggesting that PRISM learns task-discriminative multimodal representations for sentiment prediction.

For the token-level information bottleneck loss in Fig. 6(c), the curve first decreases, then temporarily increases, and finally drops again. This is reasonable because the token representations are still under-utilized at the beginning of training, and thus the bottleneck pressure can be small. When task-relevant sentiment information starts to be encoded into token representations, the bottleneck constraint becomes active, leading to the transient increase. Afterwards, the loss decreases as the bottleneck suppresses redundant token information while preserving task-relevant semantics. For the label-level information bottleneck loss in Fig. 6(d), the overall value decreases and then fluctuates within a relatively narrow range, suggesting that the label-related bottleneck consistently regularizes the learned label semantics while retaining information useful for prediction.

As for the routing loss in Fig. 6(e), its value is nearly zero during the first training stage. This is because the routing regularizer is activated only in the second stage. After the routing supervision is introduced, the loss rises temporarily, suggesting that the initial selector behavior has not yet

satisfied the routing constraint. The subsequent decline indicates that the selector gradually adapts to the regularization objective, which helps stabilize confidence-aware routing.

D Testing Stability Analysis

To assess the robustness of PRISM under random initialization, we conduct a multi-seed evaluation on CMU-MOSI and CMU-MOSEI using five seeds (42, 128, 256, 1024, 2024). All hyper-parameters are frozen at the best configuration obtained from the development set, and only the random seed is varied.

The results confirm that PRISM exhibits stable performance across random initializations. On CMU-MOSI, PRISM obtains 0.622 ± 0.020 MAE and 0.849 ± 0.008 Pearson correlation, while F1 remains above 86.8 across all five seeds (best 89.0 at seed 128). The slightly larger variance in MOSI, especially for Acc-7 ($\sigma_{\text{Acc-7}} = 2.1$), is consistent with its smaller sample size ($\sim 2,200$ samples). On CMU-MOSEI, the standard deviation of MAE is just 0.005, and the Pearson correlation remains tightly clustered at 0.799 ± 0.003 , further indicating stable behavior under training randomness.

E Dataset Composition

We evaluate PRISM on five public multimodal benchmarks covering sentiment regression, humor detection, and sarcasm detection. Table 7 summarizes the processed train/development/test splits used in our experiments.

CMU-MOSI (Zadeh et al., 2016) is a widely used utterance-level benchmark for multimodal sentiment analysis. It contains short opinion video segments with language, acoustic, and visual streams. Each segment is annotated with a continuous sentiment score in $[-3, 3]$, where negative and positive values indicate negative and positive sentiment, respectively.

CMU-MOSEI (Zadeh et al., 2018) is a larger-scale multimodal sentiment and emotion dataset collected from YouTube videos. In this work, we use the sentiment annotations, which follow the same $[-3, 3]$ scale as CMU-MOSI. The dataset provides language, acoustic, and visual modalities for each utterance-level segment.

CH-SIMS v2 (Liu et al., 2022) is a Chinese multimodal sentiment benchmark with language, acoustic, and visual modalities. Compared with CMU-MOSI and CMU-MOSEI, CH-SIMS v2 provides

fine-grained sentiment annotations in the range $[-1, 1]$. We use the multimodal sentiment labels and follow the standard train/development/test protocol.

UR-FUNNY v2 (Hasan et al., 2019) is a multimodal humor detection benchmark built from TED talk videos. Each example contains a target punchline segment and its surrounding context. The task is binary classification, where the label indicates whether the target segment is humorous. Following the HKT-aligned setting (Hasan et al., 2021), we use language, acoustic, visual, and human-centric features (HCF).

MUStARD (Castro et al., 2019) is a multimodal sarcasm detection dataset collected from TV shows and online videos. Each sample contains a target utterance together with conversational context and is labeled as sarcastic or non-sarcastic. We use the same HKT-aligned feature setting as UR-FUNNY v2, including language, acoustic, visual, and HCF modalities (Hasan et al., 2021).

For all datasets, hyper-parameters and checkpoints are selected on the development split, and the test split is used only for final evaluation.

F Introduction of baselines

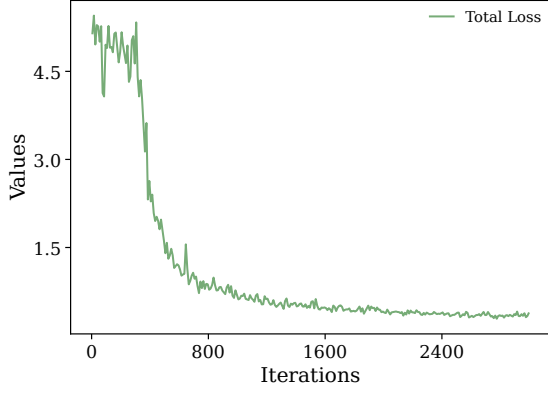
The compared baselines for MSA include:

(1) **Disentangled-Language-Focused MSA (DLF)** (Wang et al., 2025): DLF decomposes multimodal features into shared and modality-specific spaces, then uses a language-focused attractor to inject complementary acoustic and visual information into the dominant language representation.

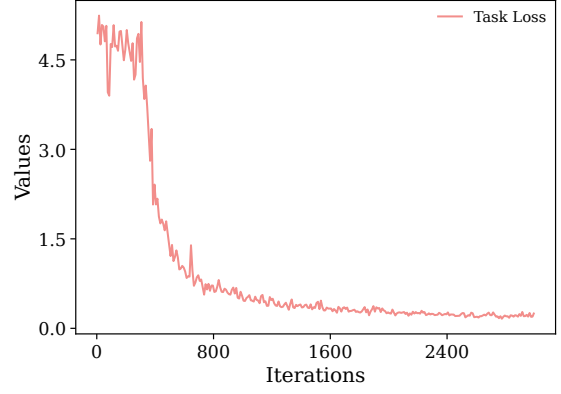
(2) **Knowledge-Guided Dynamic Modality Attention Fusion Framework (KuDA)** (Feng et al., 2024): KuDA uses sentiment knowledge to guide dynamic dominant-modality selection and adjusts modality contributions through dynamic attention fusion, enabling the model to adapt when text, audio, or vision becomes the dominant evidence source.

(3) **Emotional Description Progressive Framework (DEVA)** (Wu et al., 2025): DEVA employs an Emotional Description Generator to textualize audio and visual emotional cues, and progressively fuses these enhanced minor modalities under textual guidance to improve fine-grained sentiment recognition.

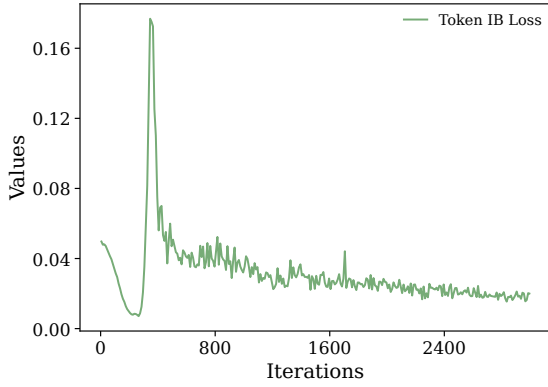
(4) **Complete Multimodal Information Bottleneck (C-MIB)** (Mai et al., 2022a): C-MIB applies the information bottleneck principle to both



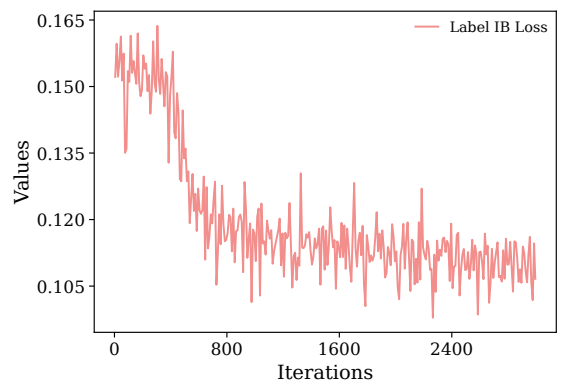
(a) Total Loss



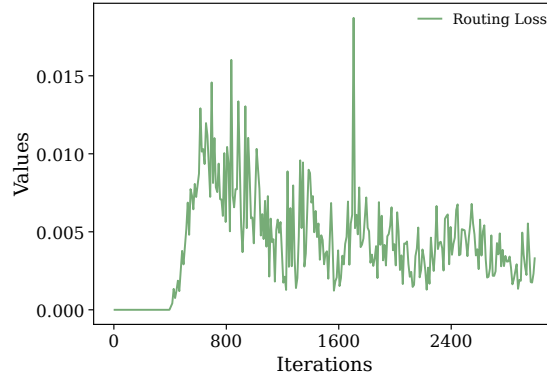
(b) Task Loss



(c) Token IB Loss



(d) Label IB Loss



(e) Routing Loss

Figure 6: Learning curves of PRISM loss components on CMU-MOSI. Each curve is averaged over every ten training iterations. The routing loss is activated only in the second training stage.

unimodal and multimodal representations, filtering noisy unimodal information before fusion and reducing redundancy in the fused multimodal embedding.

(5) **Adaptive Language-guided Multimodal Transformer (ALMT)** (Zhang et al., 2023a): ALMT learns a language-guided hyper-modality representation from acoustic and visual features, using the dominant language modality to suppress sentiment-irrelevant and conflicting information be-

fore multimodal fusion.

(6) **Information-Theoretic Hierarchical Perception (ITHP)** (Xiao et al., 2024): ITHP designates a prime modality as the input stream and uses hierarchical information bottlenecks to distill relevant information from auxiliary modalities, constructing a compact information flow for multimodal prediction.

(7) **Multimodal Boosting** (Mai et al., 2023a): Multimodal Boosting constructs multiple unimodal

Table 6: Per-seed test metrics and summary statistics (mean $\pm$ std) for five seeds.

CMU-MOSI	MAE $\downarrow$	Corr $\uparrow$	Acc-7 $\uparrow$	Acc-2 $\uparrow$	F1 $\uparrow$
seed=42	0.628	0.847	48.5	87.8	87.8
seed=128	0.606	0.856	50.8	89.0	89.0
seed=256	0.635	0.846	45.4	88.4	88.3
seed=1024	0.645	0.837	48.6	86.9	86.8
seed=2024	0.598	0.857	50.1	87.3	87.3
Mean $\pm$ Std	0.622 $\pm$ 0.020	0.849 $\pm$ 0.008	48.7 $\pm$ 2.1	87.9 $\pm$ 0.8	87.8 $\pm$ 0.9
CMU-MOSEI	MAE $\downarrow$	Corr $\uparrow$	Acc-7 $\uparrow$	Acc-2 $\uparrow$	F1 $\uparrow$
seed=42	0.500	0.800	56.1	85.2	85.4
seed=128	0.499	0.800	56.0	87.4	87.3
seed=256	0.496	0.801	56.1	87.6	87.5
seed=1024	0.509	0.800	54.4	87.4	87.3
seed=2024	0.500	0.793	55.7	86.5	86.5
Mean $\pm$ Std	0.501 $\pm$ 0.005	0.799 $\pm$ 0.003	55.7 $\pm$ 0.7	86.8 $\pm$ 1.0	86.8 $\pm$ 0.9

Table 7: Dataset statistics for the processed splits used in our experiments.

Dataset	Train	Dev.	Test	Total
CMU-MOSI	1,281	229	685	2,195
CMU-MOSEI	16,265	1,869	4,643	22,777
CH-SIMS v2	2,722	647	1,034	4,403
UR-FUNNY v2	7,614	980	994	9,588
MUSARD	539	68	68	675

and multimodal base learners, then learns contribution scores for each learner in a weakly supervised manner to reduce the influence of noisy modalities and dynamically balance modality contributions.

(8) **Disentanglement Translation Network (DTN)** (Zeng et al., 2024): DTN disentangles each modality into modality-common and modality-specific representations, then uses two-step translation and slack reconstruction to reduce distribution gaps and redundancy for multimodal sentiment analysis.

(9) **Decoupled Multimodal Distilling (DMD)** (Li et al., 2023): DMD separates multimodal features into homogeneous and heterogeneous spaces, and applies graph distillation units to learn adaptive knowledge-transfer directions and weights across modalities, reducing modality heterogeneity and enhancing discriminative representations.

(10) **Decoupled Hierarchical Multimodal Distillation (DHMD)** (Li et al., 2026): DHMD combines feature decoupling, graph-based distillation, and cross-modal dictionary matching to align heterogeneous modalities for multimodal emotion recognition.

(11) **Modality-Specific Enhanced Dynamic**

Emotion Experts (EMOE) (Fang et al., 2025): EMOE introduces a Mixture of Modality Experts to assign sample-level importance weights to language, visual, and acoustic modalities, and further uses unimodal distillation with router selection to inject modality-specific predictive information into the fused multimodal representation.

(12) **Modality Optimization and Dynamic Primary Modality Selection (MODS)** (Yang et al., 2025): MODS first reduces redundancy and noise in non-language modalities through a Graph-based Dynamic Compressor, then dynamically selects the primary modality for each sample via MSelector, and finally performs primary-modality-centric cross-attention to inject auxiliary information into the selected dominant stream.

(13) **Late-Fusion DNN (LF-DNN)** (Williams et al., 2018): LF-DNN trains separate unimodal DNN classifiers for language, acoustic, and visual modalities, and combines their prediction outputs at the decision level to obtain the final multimodal sentiment prediction.

(14) **Tensor Fusion Network (TFN)** (Zadeh et al., 2017): TFN first learns modality-specific embeddings for language, visual, and acoustic signals, then uses a tensor fusion layer to explicitly capture unimodal, bimodal, and trimodal interactions before performing sentiment prediction.

(15) **Low-rank Multimodal Fusion (LMF)** (Liu et al., 2018): LMF improves tensor-based multimodal fusion by decomposing the fusion weight tensor into modality-specific low-rank factors, enabling efficient modeling of cross-modal interactions without explicitly constructing high-dimensional multimodal tensors.

(16) **Modality-Invariant and -Specific Representations for MSA (MISA)** (Hazarika et al., 2020): MISA projects each modality into modality-invariant and modality-specific subspaces, using shared representations to reduce cross-modal distribution gaps and private representations to preserve modality-specific characteristics before multimodal fusion.

(17) **Multimodal Adaptation Gate BERT (MAG-BERT)** (Rahman et al., 2020): MAG-BERT injects acoustic and visual information into BERT through a Multimodal Adaptation Gate, which shifts lexical representations according to nonverbal cues during fine-tuning.

(18) **Self-Supervised Multi-task Multimodal sentiment analysis network (Self-MM)** (Yu et al., 2021): Self-MM uses self-supervised unimodal label generation and multi-task learning to enhance modality-specific representation learning for multimodal sentiment analysis.

(19) **MultiModal InfoMax (MMIM)** (Han et al., 2021): MMIM improves multimodal fusion by hierarchically maximizing mutual information among unimodal representations and between fused representations and unimodal inputs, helping preserve task-relevant information during fusion.

(20) **Two-stage Modality Denoising and Complementation (TMDC)** (Zhuang et al., 2026): TMDC first denoises modality-specific and modality-shared representations, then leverages the available modalities to compensate for missing ones under noisy multimodal conditions.

(21) **Diffusion Bridge** (Lee et al., 2025): Diffusion Bridge leverages a diffusion model trained on text embeddings to map vision embeddings into a text-like embedding space, reducing the modality gap between visual and textual representations for zero-shot image captioning.

The compared baselines for MHD/MSD are:

(1) **Feature-Disentangled Multimodal Emotion Recognition (FDMER)** (Yang et al., 2022): FDMER disentangles each modality into common and private representations, and fuses them with cross-modal attention for multimodal emotion recognition.

(2) **Uncertain Cross-Modal Information Bottleneck (UCMIB-PNS)** (Chen et al., 2025): UCMIB-PNS combines an uncertain cross-modal information bottleneck with probabilistic causality to reduce redundant multimodal information and dynamically balance modality sufficiency and necessity for sentiment prediction.

(3) **Attention-based Causality-Aware Fusion Network (AtCAF)** (Huang et al., 2025): AtCAF introduces causal inference into multimodal sentiment analysis, using a front-door text debiasing module to reduce textual spurious correlations and a counterfactual cross-modal attention module to filter irrelevant context during fusion.

(4) **Uncertain Cross-Modal Information Bottleneck (SuCI)** (Xu et al., 2025): SuCI mitigates subject-specific spurious correlations through causal intervention, enabling more unbiased multimodal language understanding.

(5) **MAG-XLNet** (Rahman et al., 2020): MAG-XLNet attaches a Multimodal Adaptation Gate to XLNet, using acoustic and visual cues to shift internal lexical representations during fine-tuning for multimodal sentiment analysis.

(6) **Humor Knowledge enriched Transformer (HKT)** (Hasan et al., 2021): HKT incorporates humor-centric knowledge, such as word ambiguity and sentiment cues, into transformer-based multimodal fusion to better understand humorous or sarcastic punchlines with contextual information.

(7) **Multimodal Learning using Optimal Transport (MuLOT)** (Pramanick et al., 2022): MuLOT uses self-attention to capture intra-modal dependencies and optimal transport to model cross-modal correspondence, followed by multimodal attention fusion for sarcasm and humor detection.

(8) **Multimodal Multitask Interaction Learning (MIL)** (Zhang et al., 2023b): MIL combines text-oriented multimodal attention and multitask interaction learning to jointly model sarcasm detection and sentiment analysis in conversation.

(9) **Multimodal Global Contrastive Learning (MGCL)** (Mai et al., 2023b): MGCL learns global multimodal representations through supervised contrastive learning, improving robustness for multimodal sentiment analysis and humor detection.

G Assessment Criteria

For multimodal sentiment analysis (MSA), we evaluate CMU-MOSI and CMU-MOSEI (Zadeh et al., 2016, 2018) using Acc-7, Acc-2, F1 score, MAE, and Corr. Acc-7 measures seven-class sentiment classification accuracy, where continuous predictions and labels are rounded to the nearest integer in $[-3, 3]$. Acc-2 measures binary sentiment classification accuracy after discarding neutral samples. The F1 score is the weighted F1 score under the

same binary setting. MAE denotes the mean absolute error between predictions and ground-truth sentiment scores, and Corr denotes the Pearson correlation coefficient between predictions and labels.

For CH-SIMS v2 (Liu et al., 2022), we follow its interval-binning evaluation protocol and report Acc-5, Acc-3, Acc-2, F1 score, MAE, and Corr. Predictions and labels are first clipped to $[-1, 1]$. Acc-5 is computed by assigning scores into five intervals: $(-1, -0.7]$, $(-0.7, -0.1]$, $(-0.1, 0.1]$, $(0.1, 0.7]$, and $(0.7, 1]$. Acc-3 uses three intervals: $(-1, -0.1]$, $(-0.1, 0.1]$, and $(0.1, 1]$. Acc-2 is computed by splitting scores at zero, and the F1 score is the weighted F1 score under this binary setting. MAE and Corr are computed on continuous predictions and labels.

For MHD (Hasan et al., 2019) and MSD (Castro et al., 2019), which correspond to binary humor detection and sarcasm detection tasks, respectively, we report binary classification accuracy and weighted F1 score.